

LLM-Driven Material Identification Framework

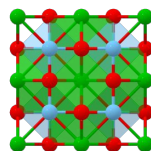
Materials Identifier

LLM-Driven Crystal Structure
Identification for DFT Simulations

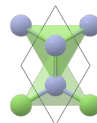
Gemini 2.5

Python 3.12

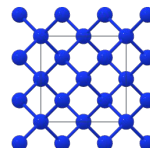
Materials Project



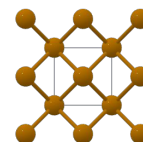
BaTiO₃



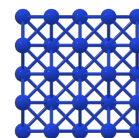
GaN



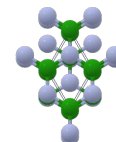
Si



Fe



Cu



hBN

Maria Minotaki

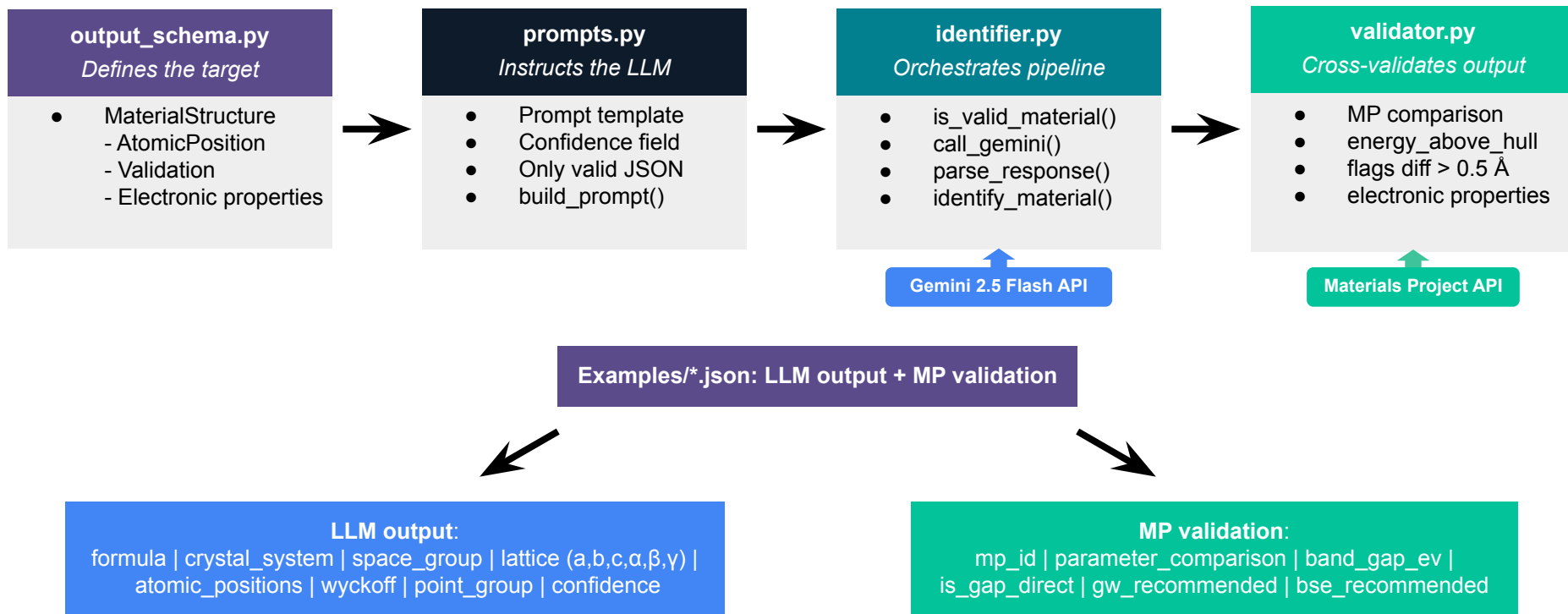
PhD, Chemical Science and Technology - ICIQ/URV

MSc, Materials Science and Technology - UoC

BSc, Physics - UoC

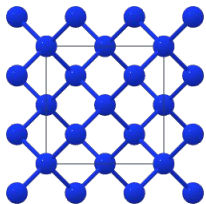
Framework Architecture

- Four modules, one pipeline: from natural language to validated DFT-ready JSON.



Observed Limitations & Future Steps

- Limitations identified from testing on 7 materials across 3 crystal systems.



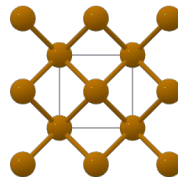
Cell Convention

Si, Cu, MgO

- Gemini: conventional cell
- MP: primitive cell



- Compare cell volumes



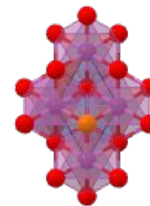
Magnetic State

Fe

- Gemini: ferromagnetic α -iron
- MP: non-magnetic entry



- Filter by magnetic state



LLM reliability

Rare & complex materials

- Hallucination undetectable
- Confidence field self-reported



- Pymatgen structure validation

Future Scope :

Richer inputs: defects, surfaces, 2D systems

Inverse design: property $\rightarrow$ structure

Thank you for your attention!

Prompt Engineering

- Where crystallography meets the language model

```
MATERIAL_IDENTIFICATION_PROMPT = """
You are an expert crystallographer and materials scientist.

RULES:
- Return ONLY valid JSON. No explanation, no markdown, no code blocks.
- All lattice parameters must be in Angstroms (a, b, c) and degrees (alpha, beta, gamma).
- Atomic positions must be in fractional coordinates (between 0 and 1).
- If you are not certain about a value, use your best scientific estimate and set confidence to "low" or "medium".
- If the material is ambiguous, identify the most common/stable form at ambient conditions.

Return this exact JSON structure:
{{
  "formula": "chemical formula e.g. Si",
  "name": "full material name e.g. Silicon",
  "crystal_system": "one of: cubic, tetragonal, orthorhombic, hexagonal, trigonal, monoclinic, triclinic",
  "space_group_symbol": "Hermann-Mauguin symbol e.g. Fd-3m",
  "space_group_number": 227,
  "point_group": "Hermann-Mauguin point group e.g. m-3m",
  "a": 5.431,
  "b": 5.431,
  "c": 5.431,
  "alpha": 90.0,
  "beta": 90.0,
  "gamma": 90.0,
  "atomic_positions": [
    {"element": "Si", "x": 0.0, "y": 0.0, "z": 0.0, "wyckoff_position": "8a"},
    {"element": "Si", "x": 0.25, "y": 0.25, "z": 0.25, "wyckoff_position": "8a"}
  ],
  "source": "LLM-inferred",
  "confidence": "high or medium or low",
  "notes": "any assumptions or caveats, or null"
}}

User description: {user_description}
"""
```

1. Role definition

“Expert Crystallographer” and “materials scientist” for improved scientific accuracy

2. JSON only

forces parseable output, not markdown

3. Explicit units & conventions

Å, degrees, fractional

4. Confidence field

Model flags its own confidence

5. Ambiguity rule

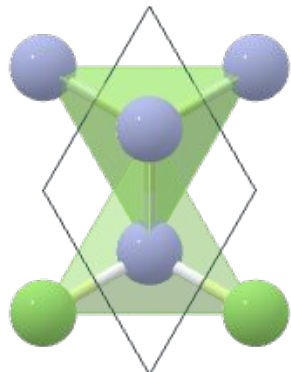
Most stable phase at ambient conditions

6. Schema as exemplar

Si example shows exact structure

Example results - GaN (Wurtzite)

- Run via CLI: `python3 identifier.py --description "wurtzite gallium nitride"`



GaN Crystal Structure
Materials Project mp-804

```
formula: GaN
name: Gallium Nitride (Wurtzite)
crystal_system: hexagonal
space_group_symbol: P63mc
space_group_number: 186
a: 3.189
b: 3.189
c: 5.185
alpha: 90
beta: 90
gamma: 120
- atomic_positions: [4 items]
source: LLM-inferred
confidence: high
notes: Lattice parameters are typical room temperature values. The internal parameter '_'
```

JSON Output Structure

atomic_positions

element:	Ga
x:	0.3333
y:	0.6667
z:	0

atomic_positions

element:	Ga
x:	0.6667
y:	0.3333
z:	0.5

atomic_positions

element:	N
x:	0.3333
y:	0.6667
z:	0.377

atomic_positions

element:	N
x:	0.6667
y:	0.3333
z:	0.877

Space Group
P6₃mc #186
Hexagonal

MP validated
 $a = 3.189 \text{ \AA}$
 $b = 3.189 \text{ \AA}$
 $c = 5.185 \text{ \AA}$

Electronic
properties
Band Gap 1.73 eV
Direct gap

The Problem

- From natural language description to DFT-ready structured output

